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Medication Management Among Older Adults Living Alone With Cognitive Impairment

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IMPORTANCE More than one-fourth of older individuals in the US with cognitive impairment live alone. These individuals often lack support for medication management and face a high risk of medication-related harm.

OBJECTIVE To elucidate barriers and facilitators to medication management experienced by older adults living alone with cognitive impairment and their social contacts.

DESIGN, SETTING, AND PARTICIPANTS This qualitative study was conducted between May 2016 and February 2024 among adults of diverse racial and ethnic backgrounds aged 55 years or older living alone with cognitive impairment in California, Louisiana, and Michigan (ie, participants), along with their social contacts, defined as family members or other familiar persons. Data were analyzed between February 2024 and March 2025.

MAIN OUTCOMES AND MEASURES As part of the Living Alone With Cognitive Impairment Project, participants' and social contacts' perspectives regarding barriers and facilitators were elicited via semistructured interviews conducted in English, Spanish, Cantonese, or Mandarin. Participants were interviewed a mean of 4 times (486 interviews total), mostly in their homes. Combined inductive and deductive thematic analysis was used to examine discussions specific to medication management.

RESULTS A total of 116 older adults living alone with cognitive impairment (median [range] age, 74 [57-103] years; 86 females [74.1%]) and 54 social contacts (median [range] age, 59 [29-89] years; 44 females [81.5%]) were interviewed. At an individual level, barriers to medication management included a lack of reminders to take medications due to a lack of cohabitants, fear of experiencing adverse effects while alone, and stress related to managing complex medication regimens with inadequate support. At an interpersonal level, barriers included difficulty in verifying medication behaviors without cohabitants, distrust of health care professionals related to a desire to maintain independence, and communication barriers. At a system level, barriers included difficulty navigating health system logistics without advocates. Participants received crucial but intermittent support with medication management from noncohabiting social contacts. Both groups suggested potential solutions to barriers, but they nevertheless conveyed a sense of precarity related to medication management.

CONCLUSIONS AND RELEVANCE This qualitative study involving racially and ethnically diverse participants from 3 states highlights the substantial barriers older adults living alone with cognitive impairment experience in managing their medications. Identification of cognitive impairment in older adults who live alone should spur targeted efforts to overcome medication-related challenges. In addition, more home- and community-based services are needed to provide medication management support to individuals living alone with cognitive impairment.

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Supplemental content

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More than one-fourth of older adults with cognitive impairment live alone in the US. This growing population of more than 4 million people is at elevated risk of self-neglect, unintentional injury, and challenges with medication management.[1-5] Medication management encompasses filling prescriptions, taking medications as prescribed, modifying regimens, and monitoring for adverse effects—cognitively challenging and error-prone processes, which become more difficult with cognitive impairment.[6,7] On average, older adults living alone with cognitive impairment receive 5 prescriptions; almost half of them are prescribed at least 1 high-risk medication, and they are more likely to both take at least 1 high-risk medication and not receive help with medication management than those living with others.[4,5,8] These individuals are at risk of underusing or misusing beneficial therapies, and/or experiencing unwitnessed adverse effects or drug-related adverse events such as falls, hospitalizations, and even death.[9-11]

Despite the high risk of adverse health outcomes faced by older adults living alone with cognitive impairment, knowledge is limited regarding their medication-related experiences.[12] Qualitative investigations of older adults with cognitive impairment in general have underscored ways in which they make sense of their medications,[13] challenges in adhering to medication regimens,[14] and the roles that caregivers play in medication management.[15] However, these studies were limited to a small number of older adults with cognitive impairment who were living alone and did not focus on medication management, precluding an in-depth understanding of unique perspectives on medication management in the context of living alone. To inform development of clinical and policy interventions to support medication management in this population, we analyzed interview data regarding barriers and facilitators to medication management from a large and racially and ethnically diverse group of older adults living alone with cognitive impairment and their social contacts.

Key Points

Question What are the barriers and facilitators to medication management experienced by older adults living alone with cognitive impairment and their social contacts?

Findings In this qualitative study of 116 adults aged 55 years or older living alone with cognitive impairment and 54 social contacts, participants described unique barriers to managing medications while living alone, in particular, lacking medication reminders, fearing adverse effects when alone, withholding information from health care professionals to maintain independence, and encountering logistical difficulties without advocates.

Meaning Detecting cognitive impairment and medication management challenges in older adults living alone is critical to achieving safe and effective medication use in this population.

Methods

This qualitative study was approved by the review board at the University of California, San Francisco, and followed the Consolidated Criteria for Reporting Qualitative Research (COREQ) guidelines; older adults and social contacts provided informed consent, including permission to publish quotations.[16]

Study Design

The parent project of this study was the Living Alone With Cognitive Impairment Project, a mixed-methods investigation of the barriers and facilitators to accessing essential services experienced by racially and ethnically diverse older adults living alone with cognitive impairment in the US (eMethods in Supplement 1).[17] To ensure a comprehensive perspective, the parent project was designed to draw from a socioecological model to identify barriers and facilitators at the individual, interpersonal and/or relational, and system and/or policy levels.[18] This article describes issues related to medication management identified through longitudinal interviews regarding access to essential services in the qualitative arm of the mixed-methods parent project. The large sample size of the qualitative arm was established to assess the transferability of findings originating from the urban San Francisco Bay Area to rural areas and other states.[19-22]

Participant Recruitment and Data Collection

We purposively sampled individuals to represent diverse sex, racial and ethnic backgrounds, and states of residence using a variety of recruitment strategies (eMethods in Supplement 1). The primary source of recruitment was introductions from trusted sources, such as physicians, social workers, and community gatekeepers. Race and ethnicity were included in the analysis to reflect the diversity of the US population and account for elevated rates of cognitive impairment, disproportionately undiagnosed, among minority communities.[23,24] Self-reported racial and ethnic categories were as follows: Asian, Black or African American, Hispanic, Native American, White, and other or preferred not to answer. Inclusion criteria for older adults living alone with cognitive impairment (ie, participants) were age 55 years or older; living alone (defined as spending the night alone most weeknights); having a physician-, self-, or informant-reported diagnosis of mild cognitive impairment, Alzheimer disease, or non-Alzheimer dementia or scoring 24 or lower on the Montreal Cognitive Assessment; and being able to provide informed consent. Exclusion criteria included residing in an institution (eg, assisted living facility) and having a severe psychiatric disorder. We invited participants to provide contact information for 1 family member or other familiar person who might be interested in participating in the study. Inclusion criteria for social contacts included an age of 18 years or older and being referred by a study participant with cognitive impairment.

Semistructured interview guides (eAppendix in Supplement 1) for older adults living alone with cognitive impairment and social contacts were developed by our multidisciplinary team, including community advisors.[25] A cognitive neuroscientist (J.K.J.) ensured that the interview guide was appropriate for persons with cognitive impairment, which required using simple questions, speaking slowly, and using silence to support information processing.[26]

Table 1. Self-Reported Demographics of Older Adults Living Alone With Cognitive Impairment and Their Social Contacts

Characteristic	No. (%) Older adults living alone with cognitive impairment (n = 116)	No. (%) Social contacts (n = 54)
Age, median (range), y	74 (57-103)	59 (29-89)
Sex		
Female	86 (74.1)	44 (81.5)
Male	30 (25.9)	10 (18.5)
Race		
Asian	29 (25.0)	19 (35.2)
Black or African American	28 (24.1)	10 (18.5)
Native American	1 (0.9)	1 (1.9)
White	37 (31.9)	17 (31.5)
Other or not provided[a]	21 (18.1)	7 (13.0)
Ethnicity		
Hispanic	24 (20.7)	6 (11.1)
Non-Hispanic	84 (72.4)	41 (75.9)
Not provided	8 (6.9)	7 (13.0)
Educational attainment		
Less than high school	45 (38.8)	4 (7.4)
High school; no higher degree	35 (30.2)	12 (22.2)
Associate's degree or higher	35 (30.2)	31 (57.4)
Not provided	1 (0.9)	7 (13.0)
Medicaid insurance	68 (58.6)	NA
Marital status		
Widowed	46 (39.7)	5 (9.3)
Divorced	31 (26.7)	7 (13.0)
Never married	19 (16.4)	14 (25.9)
Married or partnered	1 (0.9)	19 (35.2)
Separated	13 (11.2)	0
Not provided	6 (5.2)	9 (16.7)
Country of birth		
US	65 (56.0)	32 (59.3)
China	24 (20.8)	18 (33.2)
Mexico	7 (6.0)	0
El Salvador	5 (4.3)	1 (1.9)
Other[b]	15 (12.9)	3 (5.6)
Language of choice		
English	66 (56.9)	34 (63.0)
Spanish	21 (18.1)	2 (3.7)
Cantonese	9 (7.8)	2 (3.7)
Mandarin	17 (14.7)	16 (29.6)
Other[c]	3 (2.6)	0

(continued)

Table 1. Self-Reported Demographics of Older Adults Living Alone With Cognitive Impairment and Their Social Contacts (continued)

Characteristic	No. (%) Older adults living alone with cognitive impairment (n = 116)	No. (%) Social contacts (n = 54)
State or country of residence		
California	94 (81)	30 (55.6)
Michigan	15 (12.9)	14 (25.9)
Louisiana	8 (6.9)	7 (13.0)
Mississippi	0	1 (1.9)
Florida	0	1 (1.9)
Peru	0	1 (1.9)
Geographic location		
Urban	84 (72.4)	33 (61.1)
Suburban	14 (12.1)	15 (27.8)
Rural	18 (15.5)	6 (11.1)
Living arrangement		
Apartment complex	33 (28.4)	NA
Single family home	31 (26.7)	NA
Senior housing	42 (36.2)	NA
Single occupancy room	8 (6.9)	NA
Other[d]	2 (1.7)	NA
Cognitive impairment diagnosed		
No	76 (65.5)	NA
Yes	40 (34.5)	NA
Montreal Cognitive Assessment score (among undiagnosed subsample; n = 74), mean (SD)[e]	17.1 (5.3)	NA
Chronic health conditions		
Hypertension	42 (36.2)	NA
Diabetes or high blood glucose	19 (16.4)	NA
Cancer	14 (12.1)	NA
Psychiatric problem	20 (17.2)	NA
Heart problem	24 (20.7)	NA
Stroke	15 (12.9)	NA
Health status		
Poor	21 (18.1)	NA
Fair	48 (41.4)	NA
Good	24 (20.7)	NA
Very good	18 (15.5)	NA
Excellent	1 (0.9)	NA
Not provided	4 (3.4)	NA

(continued)

Interviews and observations were conducted in English, Spanish, Cantonese, or Mandarin between May 2016 and February 2024 (eMethods in Supplement 1). Interviews were audio recorded and transcribed verbatim. To build trust, our team generally interviewed older adults living alone with cognitive impairment 4 times, usually in their homes, over a 6-month period; we interviewed social contacts once. Older adults living alone and social contacts were usually interviewed separately to ensure that we elicited their unique perspectives. During the first interview, older adults living alone with cognitive impairment were asked what medications they used and how they managed their use. Subsequent interviews focused on other topics but elicited updates regarding medication management as relevant. Researchers supplemented interviews with observations of participants' environments, with field notes to add context (eg, documenting data not captured by audio recordings, such as observing pills on the floor).

Data Analysis

Interviews were translated to English for analysis. Interview transcripts and field notes were uploaded into ATLAS.ti, version 25.0.2 (Scientific Software Development GmbH), and analyzed using combined inductive and deductive thematic analysis.[27-29] Informed by the socioecological model,[18] between February 2024 and March 2025, a team of 4 coders supervised by E.P. and T.T. created codes to identify barriers and facilitators to managing medications while living alone with cognitive impairment (eMethods in Supplement 1). Transcripts including field notes were independently coded by 1 coder and reviewed by another for consistency. Disagreements between coders were resolved through discussion until consensus was reached.

M.E.G. and E.P. identified themes related to barriers and facilitators to managing medications while living alone with cognitive impairment by inferring relationships between codes. Emerging relationships were elucidated by creating tables in Microsoft Excel (version 16.1), diagramming, and holding iterative discussions with the study team, which facilitated the cohesive description of dynamics emerging from the data.[29] M.E.G. and E.P. subsequently categorized themes regarding barriers and facilitators to medication management into individual, interpersonal and/or relational, and system and/or policy levels.[18]

Table 1. Self-Reported Demographics of Older Adults Living Alone With Cognitive Impairment and Their Social Contacts (continued)

Characteristic	No. (%) Older adults living alone with cognitive impairment (n = 116)	Social contacts (n = 54)
No. of medications and/or supplements taken by participant (among those with complete counts; n = 107), mean (SD)[f]	5.1 (3.6)	NA
Relationship to person with cognitive impairment		
Daughter	NA	24 (44.4)
Friend	NA	17 (31.5)
Son	NA	4 (7.4)
Case manager	NA	2 (3.7)
Home care aide	NA	2 (3.7)
Sister	NA	1 (1.9)
Romantic partner	NA	1 (1.9)
Son-in-law	NA	1 (1.9)
Cousin	NA	1 (1.9)
Granddaughter	NA	1 (1.9)
Social contact lives within 10 mi of older adult	NA	32 (59.3)

Abbreviation: NA, not applicable.

[a] Twenty of 21 older adults living alone with cognitive impairment chose other for race and primarily identified as Hispanic. Five of 7 social contacts similarly chose other for race and primarily identified as Hispanic. One of 21 older adults living alone with cognitive impairment and 2 of 7 social contacts did not provide a self-identified race.

[b] For older adults living alone with cognitive impairment, other included Colombia, Cuba, Denmark, Ecuador, Guatemala, Hong Kong, India, Jamaica, Nicaragua, Peru, Puerto Rico, former Soviet Union, and Vietnam. For social contacts, other included Ecuador, Guatemala, and Peru.

[c] For language of choice, other included Jamaican Patois, Japanese, and Russian.

[d] One participant resided in a remote cabin followed by a truck in a rural area, and another participant lived alone in a room while often changing residence.

[e] The Montreal Cognitive Assessment is scored out of 30. Cognitive testing was incomplete for 2 participants.

[f] Complete medication counts were unavailable for 9 participants.

Results

A total of 116 older adults living alone with cognitive impairment (median [range] age, 74 [57-103] years; 86 females [74.1%]) and 54 social contacts (median [range] age, 59 [29-89] years; 44 females [81.5%]) participated in the study (**Table 1**). We interviewed participants a mean of 4 times (486 interviews total; mean [range] interview duration, 62 [20-165] minutes); social contacts were usually interviewed once (mean [range] interview duration, 48 [26-154] minutes). Among older adults living alone with cognitive impairment, 29 (25.0%) self-identified as Asian, 28 (24.1%) as Black or African American, 24 (20.7%) as Hispanic, 1 (0.9%) as Native American, and 37 (31.9%) as White; 21 (18.1%) reported their race as other or preferred not to answer. Most social contacts were family members.

We organized emergent themes into individual, interpersonal and/or relational, and system and/or policy levels (**Figure**). **Table 2** shows representative quotations regarding barriers to medication management related to living alone. Participant insights into potential solutions for medication management barriers are listed in **Table 3**. Potential solutions to overcome barriers to medication management and clinical and/or policy implications developed by the study team based on study findings and the literature are shown in **Table 4**.[11,30-58]

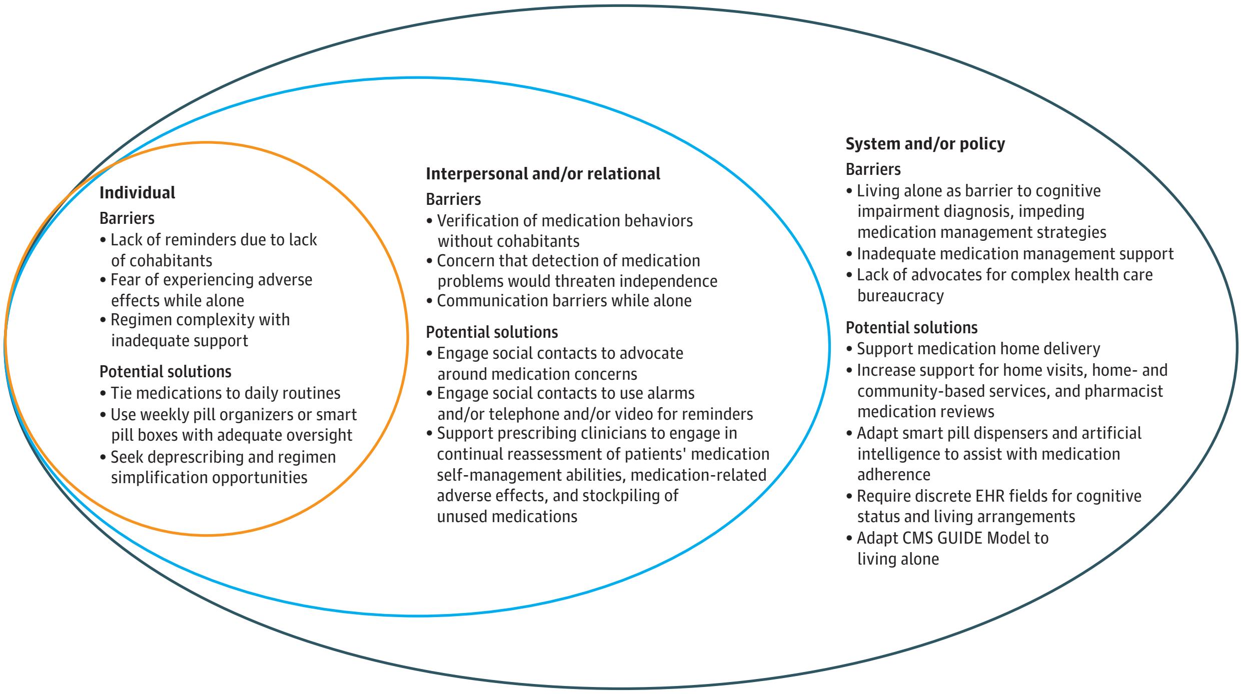
Individual-Level Themes

Older adults living alone with cognitive impairment experienced daily struggles to remember and manage complex medication regimens, cope with fears of adverse effects occurring while alone, understand prescriptions, and devise strategies to overcome these barriers.

Lack of Reminders to Take Medications Due to Lack of Cohabitants

Older adults frequently cited memory impairment in the context of living alone as a barrier to taking medications as prescribed. Participants emphasized being vulnerable to forgetting medications during periods of time when they were alone. One participant said, "Nobody pushes me. There's nobody telling me, 'Did you take your pills already? Come on, take them.'" These concerns were echoed by social contacts, who worried whether participants were able to accurately report their ability to take medications in the context of living alone. One woman in her 50s described how her mother, who is in her 80s, "doesn't remember a thing anymore. Sometimes, she says she has already taken the drug when she hasn't. Sometimes, she says

Figure. Conceptual Diagram of Study Findings



Barriers to medication management raised by older adults living alone with cognitive impairment and social contacts, as well as potential solutions raised by these 2 groups and the study team. CMS indicates Centers for Medicare & Medicaid Services; EHR, electronic health record; GUIDE, Guiding an Improved Dementia Experience. The concentric nesting of barriers and solutions depicts the interconnected nature of multilevel barriers as well as the potential to intervene on barriers at multiple levels.

she hasn't taken the drug when she has already taken them. Therefore, it is always messed up."

Participants and social contacts deployed myriad strategies to help participants remember medications during unsupervised periods, including posting reminders around their homes, filling pill organizers, and linking medication-taking behaviors to routines, such as brewing coffee or daily prayers. Older adults living alone with cognitive impairment explained that long-standing use of medications facilitated remembering medications. Nevertheless, many participants remained apprehensive regarding their ability to remember taking medications or related tasks (eg, rotating insulin injection sites). Some social contacts attempted to monitor medication-taking behaviors, including by frequent telephone check-ins, as described by a woman in her 50s: "'Dad, I'm not hanging up until you take them.' Because he would get distracted and forget."

Without regular companions to demystify knowledge gaps, older adults living alone with cognitive impairment exhibited varying degrees of understanding their prescribed medications. Challenges included difficulties recalling medication names, the rationale for each medication, potential adverse effects, and their prescribers. Asked about a medication, a female participant in her 70s commented, "I don't know what it is, or what it's for, or anything." In some cases, social contacts filled in medication details, contextualizing the rationale for medications, and accompanied participants to appointments to track changes and report adverse effects.

Fear of Adverse Effects Occurring While Alone and Being Unable to Get Help

Older adults living alone with cognitive impairment and social contacts frequently raised concerns about potential and previously experienced adverse effects of prescribed medications and feared that drug-related adverse effects would arise during times when older adults were alone. Of particular concern were adverse effects that might cause participants to feel drowsy or confused. Another female participant in her 70s noted, "I took half a pill and slept for half a day. I barely ate anything. I basically took a shower and then fell asleep."

Stress Related to Medication Regimen Complexity With Inadequate Support

Participants' challenges in remembering medications and fear of adverse effects were exacerbated by complex medication regimens. Participants described being burdened by polypharmacy. Others contended on their own with self-monitoring challenges, such as performing finger sticks to check blood glucose in the setting of tremors. Managing the timing of medications alone was challenging. Participants cited difficulties handling small pills while alone, which they struggled to place

Table 2. Barriers to Medication Management Specifically Related to Living Alone

Level of socioecological model and barrier specifically related to living alone	Quote
Individual	
Lack of reminders to take medications due to the lack of cohabitants	"[It's] just me. I don't have anybody to tell me, 'Let me put some cream on you. I'll help you.' No, just me." —Female participant in her 70s
	"If she [forgets her medications] 1 night I'm not here to remind her because I'm not here in the evening." —Granddaughter of a female participant in her 90s
Fear of adverse effects occurring while alone and being unable to get help	"Sometimes I talk to her and she's totally gone [confused]. She lives at home alone. She shouldn't be this medicated because she is not here." —Daughter of a female participant in her 70s
	"[My friend] got diagnosed with prostate cancer and was going to need some help for a couple of days following that procedure. In the course of helping him around that, I realized that I needed to be there when he was on pain medications and had the effects of the anesthesia and stuff, that it wasn't really safe to leave him alone with medications, in cooking and things during that time." —Female friend of a male participant in his 60s
Stress related to medication regimen complexity with inadequate support	"The bad part of [the nebulizer] is that, so you hook all that jazz up, okay? This goes to this and then you turn the machine on, it blows, and then you put the medicine in here and it's like, it makes air kind of, misty air.... Well, these little things, I'm supposed to do 2 a day, 3 a day actually, but look at how little it is. And for what's in there, it takes forever to do it." —Female participant in her 70s
	"My machine for checking my blood sugar doesn't work. It has been a week, and it doesn't work. I should check myself in the morning and at night. I don't know what's wrong with it. [At the community clinic] from the second floor, they send me to the third. From the third, to the second. No, I don't know what's happening, and I just came home instead. I don't like it." —Male participant in his 60s
Interpersonal and/or relational	
Challenges in verifying medication behaviors and self-reports in the absence of cohabitants	"[My brother] is on insulin. And at a certain point, it's going to cause problems because he doesn't want anybody to see him shoot, and he'll forget to do it." —Sister of a male participant in his 70s
	"I took her answers exactly as she said it, and it's clear that the answers do not really portray the reality that she needs help with medication." —Field note from interview with a female participant in her 70s
Distrust and concern that detection of medication-related problems would threaten ability to live alone	"The psychiatrist wants me to take the medications, but oftentimes I'm afraid to take them. I would lie to the doctor that I've taken them. The psychiatrist would be okay with that. It's okay for the doctor as long as I'm taking the medications.... If he knows about my situation, he would call 911 and put me into an inpatient psychiatric ward." —Male participant in his 70s
Communication barriers and perceived lack of explanation regarding medications	"[At the pharmacy] I probably understand 1 or 2 sentences that they say in English. That's all. Just understand a little. I understand the simple ones, but as for the complicated ones, I have no idea what they are saying." —Female participant in her 70s who primarily speaks Mandarin
	"The side effects [of gabapentin] are not good at all for me. And it says [on the drug label], 'if you notice other effects not listed, contact your doctor or pharmacist.' And you think they'd know something like this could be happening? And my doctor sure knows that it does. So, it seems like they would take more interest and look, find out why and how and what it's doing to me. But they don't seem to be interested." —Female participant in her 80s
System and/or policy	
Living alone as a barrier to being diagnosed with cognitive impairment, impeding proper medication management strategies	"But even though she's in a program that is supposed to support her...it seems to me that (1) they didn't tell her that she's got a diagnosis of cognitive impairment, whatever it is [and] (2) they're not really supporting her because there is this...home care aide that is going only twice a week, and she's just barely cleaning and doing the dishes but is not really getting more help. And [the participant] doesn't have somebody helping with the finances, she doesn't have somebody helping with the groceries, she doesn't have anybody helping with the laundry. So she's trying to wash her blankets, manage her pills. They're just calling her to ask if she took the pills." —Field note from interview with a female participant in her 70s
Inadequate medication management support systems	"I have all this medicine stockpiled all over the... house because they keep sending me medicine. I'm supposed to be using it." —Female participant in her 90s
	"She [the physician] just told me, 'Take them [pills]. Put them out in front of you so that you remember.'... 'Doctor, I'm forgetting my pills,' I told her, too. 'No, no, no,' she told me. 'You can't even stop taking your medicine for 1 day. You have to take them daily." —Female participant in her 70s
Lack of advocates to help with complex health care bureaucracy	"It is Medi-Cal, it's a struggle. I had to go down to Medi-Cal [Medicaid office] yesterday because I, me, with my organization, don't want to deal with that. Because when you go down there it's a headache. It's like you see so many people struggling, and I see myself in all those people because I'm going to be there." —Female participant in her 60s
	"The care is pretty good [at a local older adult care program], but I think her not having a family member or someone to advocate for her, that maybe they need a little push sometimes to help her." —Friend of a female participant in her 70s

in and remove from pill organizers. Bubble packs also led to dropping pills on the floor and subsequently struggling to recognize small pills. Social contacts and home aides provided hands-on support to offer medications, supervise administration, and offload medication-related tasks; however, such intensive support was often inadequate.

Interpersonal- and/or Relational-Level Themes

The inability of noncohabiting social contacts and study team members to observe all aspects of medication-related behaviors heightened the sense of uncertainty surrounding medication management. Noncohabiting social contacts had difficulty obtaining reliable information regarding

Table 3. Potential Solutions to Medication Management Barriers for Older Adults Living Alone: Participant-Reported Insights

Level of socioecological model and insights	Quote
Individual	
Tying medication-taking behaviors to daily routines was critical to remembering to take medications while alone.	"I find that I forget. So I have…a piece of white paper and put on it, 'take pills,' and when I make the bed every morning I put it on the pillow. And so when I go in there, if I haven't taken it off the pillow then I haven't taken the pills. So, I come in and take them because I don't want to forget." —Female participant in her 70s
	"We set up…a daily to-do list for him to use when I wasn't there, and it's sort of expanded into what he calls his daily recovery schedule…. That's going to help him maintain his independence the longest. Because as long as he follows that, he doesn't forget to take his meds or to check his glucose, big deal things, that with as many meds as he takes and as many diagnoses as he has, could end up messing someone else's life up." —Female friend of a male participant in his 60s
Weekly pill organizers, including smart pill dispensers, blister packs, and electronic medication dispensers, were generally helpful but worsened confusion when not used correctly or with inadequate oversight from social contacts.	"I tell Alexa [Amazon device] to make sure I take my medicine at 6:30 AM. 'Alexa, tell me.' And at night, I tell Alexa, 'Don't forget to remind me to take my other medicine at 8:00.' Alexa remembers…. I rely on Alexa and my phone a lot." —Female participant in her 70s
	"We now mainly watch her to take this antihypertensive medicine because she will feel dizzy if she forgets to take it. We help her to put the medicine in the medicine box. We check her medicine box and help her organize it. She basically knows these medicines because she takes them frequently. When she is at my house, it is okay because I am by her side. Sometimes she forgets when she goes back to the apartment by herself." —Daughter of a female participant in her 80s
Interpersonal and/or relational	
In some cases, social contacts played a pivotal role in helping older adults living alone with cognitive impairment raise medication-related concerns to providers.	"When I saw the urologist, I thought that since it was the medicine prescribed by the doctor, I must take it…. My blood pressure was already low, so after taking it, it became even lower, which was more dangerous. So, I asked my friend to write a general description of my condition and let my doctor see it, so they stopped the medicine for me. I didn't know that taking the medicine would cause so many things to happen." —Female participant in her 70s
	"We found out that, by [my mother] getting with different doctors for different situations, [her physicians] provided her medication without realizing that she was already taking another type of medication. So [my sisters, brothers, and I] got with her primary care doctor, and they took away a lot of the medicine, because a lot of it was having her just feeling like bad, down. So since they've corrected that she's seemed to be much better." —Daughter of a female participant in her 80s
Social contacts set up alarms for older adults living alone and/or video-chatted with them to ensure adherence.	"The alarm goes off when I'm supposed to take [my pills]. I just go swallow them. But there is an alarm…. It is a little clock in the bottom that says, 'Take your pills,' and so the alarm is set 4 times a day, and at that alarm, I take those pills. My daughter brought it." —Female participant in her 70s
System and/or policy	
Medication home delivery was often considered helpful but, in a subset of participants, led to stockpiling when the participant was not taking medications.	"I get all my prescriptions delivered now, so I don't have to go to the pharmacy…. That was frustrating because I'd walk to the pharmacy, and the drugs wouldn't be ready, and the lines would be long. So, now having them delivered is much nicer." —Female participant in her 70s
If possible, home visits gave health care professionals an opportunity to detect and address medication-related problems through direct observation.	"[The study participant living alone with cognitive impairment] had a lot of diarrhea, and he couldn't control it. And so he had diarrhea on the floor, and the [home care physician] cleaned it. And [the home care physician] was concerned about it. And then the doctor realized that, looking at the medicines, that the medicines for diarrhea were missing in the small boxes that [the participant] had…. Since the pill for diarrhea was missing he had this diarrhea. But now it is fixed because [the home care physician] put back the diarrhea pills…. That's the value of having…doctor home visits, because she can check what's going on. Doctor home visits are really valuable, also because for this man it's really hard to go to the doctor." —Field note from interview with a male participant in his 60s
Some participants received critical support with medication management from home- and community-based service professionals.	"My friend went to get trained with In-Home Supportive Services, and so she supervises me. Probably every 2 weeks she'll come because I have 2 big pill boxes…. She has a nice technique to make sure I put them in there correctly." —Male participant in his 60s
	"An organization that can come in and fill the pill containers for somebody is a big deal…. If you make a mistake on 1 dose, that's 1 thing, but if you skip your heart medicine for a whole week, that's another thing." —Daughter of male participant in his 80s

medication management. Interactions between health care professionals and older adults living alone with cognitive impairment were hampered by pervasive distrust and communication barriers.

Challenges in Verifying Medication Behaviors and Self-Reports in the Absence of Cohabitants

Without directly observing the taking of medications, social contacts relied on the self-report of participants and often described finding pill organizers with unused pills after periods when participants had been alone. Many were left with a partial understanding of the medication-taking behaviors. A daughter in her 50s doubted her mother's (female participant in her 80s) self-report: "You had no idea whether she took [pills] or not…. I discovered a lot of pills at [her] home." Study team members similarly experienced uncertainty regarding the precise details of medications taken by participants, with extended exchanges sorting through medication bottles and attempts to elucidate medication regimens.

Table 4. Proposed Medication Management Strategies Synthesized From Study Findings and Literature

Barrier related to living alone with cognitive impairment	Potential solution
Individual	
Lack of reminders	Initiate counseling and/or brainstorming between patients, social contacts, and health care professionals regarding ways to incorporate medication taking into routines[a,30-32]
	Use pill boxes and/or organizers[a,33]
	Continually assess patients' ability to self-manage medications (eg, by inquiring about medication challenges or mistakes in a nonjudgmental fashion, or via direct observation/structured assessment) along with availability and quality of in-home support[11,34]
	Adapt emerging technologies such as smart pill dispensers to unique needs of older adults living alone with cognitive impairment[35,36]
	Use home delivery of medications and automated refills[a,37]
	Encourage health care professionals and social contacts to inquire about stockpiling of medications (eg, "Do you have unused or expired medications at home?")[32,38,39]
Adverse effects	Assess and address barriers to adherence, including potential or experienced drug-related adverse effects, during all health care encounters[a,34]
	Leverage emerging technologies and artificial intelligence applications to rapidly detect dangerous adverse effects[35,40]
Regimen complexity	Seek out opportunities for deprescribing of potentially harmful and/or unnecessary medications and regimen simplification[a,11,41]
Interpersonal and/or relational	
Verification of medication behaviors	Assess patients' eligibility and expand opportunities for home visits, residential living options, or PACE[a,42-44]
	Adapt emerging technologies (eg, voice-assisted or connected home devices) to augment oversight of medication-taking behavior and to enable linkages between devices and clinical records[45]
Distrust and communication	Promote longitudinal, patient-oriented care, with consideration for how prescribing decisions may affect patients' independence[a,46]
	Include family and caregivers, including trusted nonkin friends and neighbors, in clinical encounters and medical decision-making[a,46]
System and/or policy	
Delayed diagnosis of cognitive impairment	Consider targeted cognitive impairment screening among older adults who live alone[47]
	Change electronic health records so that living arrangement can be systematically tracked, particularly for those with identified cognitive impairment[48-50]
Inadequate support systems	Bolster and expand Medicaid eligibility and access[43,51]
	Prioritize recruitment and retention of workforce with robust training in medication management[43]
	Support pharmacist-led interventions aiming to improve prescribing quality and adherence[52,53]
Lack of advocates	Use care navigation and dementia-specific care coordination[a,54]
	Adapt CMS GUIDE Model to those living alone[54-57]
	Leverage and adapt emerging technologies and artificial intelligence applications to support older adults living alone with cognitive impairment with care navigation[58]

Abbreviations: CMS, Centers for Medicare & Medicaid Services; GUIDE, Guiding an Improved Dementia Experience; PACE, Program of All-Inclusive Care for the Elderly.

[a] These potential solutions emerged directly from insights from older adults living alone with cognitive impairment and/or their social contacts.

Distrust and Concern That Detection of Medication-Related Problems Would Threaten Ability to Live Alone

While some older adults living alone with cognitive impairment trusted professionals involved in medication prescribing and administration, many distrusted health care professionals. Distrust often stemmed from concern that the detection of medication-related problems may threaten their ability to live alone. Some participants shared that they intentionally withheld information from health care professionals or were not truthful in reporting adherence. A male participant in his 70s said, "My doctor asks me to take them. I just reply I have taken them already. I lie to my doctor that I've taken them already. I threw the medications away the next second." A female participant in her 80s reported that a home aide "took pictures of a pill cup with pills in it to say I'm not taking my prescriptions.... I think she's trying to say that I don't take my pills, so I belong in a nursing home."

Distrust also resulted from feeling that health care professionals had prescribed an excessive number of medications and from previously experienced adverse effects in the context of living alone. Generally, participants trusted their social contacts more than their health care professionals. Social contacts leveraged this trust to encourage participants to speak more transparently with health care professionals about medication-related problems.

Communication Barriers and Perceived Lack of Explanation Regarding Medications

Participants often faced substantial communication barriers with health care professionals, hindering their ability to understand medication-related instructions on their own. In 50 of 116 participants (43.1%) who had limited English proficiency, this stemmed from language discordance. A female participant in her 70s said, "I don't understand English. They didn't tell me what not to do or what to pay attention to or what reactions would happen. If there is a reaction, what should I do?" Social contacts, if available, helped translate instructions and facilitate discussions with prescribers.

System- and/or Policy-Level Themes

Systemic difficulties related to medication management included lacking a diagnosis of cognitive impairment while living alone and having inadequate medication management support systems. These challenges were exacerbated by a lack of advocates to help navigate complex health bureaucracies.

Living Alone as a Barrier to Being Diagnosed With Cognitive Impairment, Impeding Proper Medication Management Strategies

Almost two-thirds of participants (76 of 116 [65.5%]) reported not having received a cognitive impairment diagnosis despite meeting eligibility criteria based on cognitive screening. One consequence of receiving such a diagnosis was that social contacts, when available, were alerted to the importance of assisting participants with medication management. However, specific arrangements initiated by prescribers to accommodate participants based on their status of living alone with diagnosed cognitive impairment did not emerge. Additionally, we did not note substantial differences in barriers to medication management or proposed solutions in older adults living alone with cognitive impairment who were diagnosed compared with those who reported being undiagnosed.

Inadequate Medication Management Support Systems

Participants described system-based solutions aiming to promote medication adherence, such as home medication delivery, pill organizers or medication blister packs, and electronic medication dispensers (Table 3). For example, a female participant in her 70s described using a smart device to remind her to take her medication at specific times of day. However, participants continued to experience logistical difficulties despite these solutions. For example, several participants described stockpiling unused mail-delivery medications around their homes (Table 2); others were confused by electronic pill dispensers.

Lack of Advocates to Help With Complex Health Care Bureaucracies

Older adults living alone with cognitive impairment described difficulty navigating complex health care bureaucracies in which they felt overlooked. Common refrains included receiving confusing letters regarding unexpected changes in drug coverage benefits, spending hours on impenetrable telephone systems, facing long response times, and feeling disempowered to contact health care professionals or insurers. Social contacts, if available, were essential advocates in navigating the bureaucracy. They also helped in communicating with health care professionals and pharmacies regarding refills, picking up prescriptions, and providing transportation to and/or from the pharmacy. Even amid many barriers, participants were often grateful for insurance coverage through Medicare and/or Medicaid. Nevertheless, they expressed concern regarding the expense of care on a single-householder budget; according to a female participant in her 70s, the "cost burden is quite heavy."

Discussion

Older adults with cognitive impairment and their social contacts highlighted significant barriers to medication management, which were exacerbated by living alone and spending large amounts of time unaccompanied. Our findings align with prior research showing that individual, clinical, social, and environmental factors interact to influence the medication adherence of older adults.[7,11,59] Specific to living alone, participants described daily struggles to remember medications without external reminders or encouragement, fearing adverse effects (particularly drowsiness) while alone, withholding medication-related information to maintain their independence, and navigating medication-related bureaucracies with limited accompaniment. While participants received crucial but inconsistent support from noncohabiting social contacts, both groups conveyed a sense of precarity related to medication management activities.[2] Nevertheless, older adults living alone with cognitive impairment and their social contacts shared insights regarding potential solutions to overcome medication management barriers.

A growing evidence base supports the health-related vulnerabilities of older adults living alone with cognitive impairment[1,2,60]—a population that is growing due to increased longevity,[61,62] divorce rates,[63] and childlessness.[64] Prior work has underscored that older adults living alone with cognitive impairment have inadequate access to essential services, including insufficient support with medication management.[3-5] With a qualitative approach, we elucidate the lived experiences of managing medications in this underserved population. Study findings additionally illuminate guideposts for developing interventions to improve medication management for older adults living alone with cognitive impairment.

We identified potential solutions to overcome barriers to medication management, along with clinical and/or policy implications, based on study findings and the literature. For example, at the individual level, a frequently cited barrier to medication adherence was the fear of experiencing drug-related adverse effects while being unaccompanied. Accordingly, when caring for older adults living alone with cognitive impairment, health care professionals and health systems should prioritize assessing and addressing concerns regarding potential and/or experienced drug-related adverse effects at health care encounters, adopting a nonjudgmental approach that emphasizes how common these may be.[65] Technological solutions already being deployed among patients living alone with cognitive impairment and emerging technologies, such as artificial intelligence, should be adapted to this population. For example, smart pill dispensers checked remotely by health care professionals should be explored as a means to rapidly detect dangerous adverse effects occurring while patients are alone and alert trusted contacts and/or health care practitioners.[35]

At the interpersonal and/or relational level, participants voiced concern that detection of medication nonadherence would threaten their ability to live in their home. This finding supports the importance of longitudinal, patient-oriented care in which health care professionals attend to the feelings, perspectives, and beliefs underpinning medication use.[59,66-68] Similarly, strategies must facilitate communication and trust between patients living alone and health care professionals so patients living alone are comfortable raising medication-related concerns.[68,69]

At the system and/or policy level, living alone emerged as a barrier to being diagnosed with cognitive impairment and/or being aware of such a diagnosis. Medication management difficulties, including nonadherence, are among the first signs of cognitive impairment and should prompt a cognitive assessment.[70-72] Almost two-thirds of participants reported being undiagnosed despite having cognitive impairment by cognitive testing, mirroring national rates and representing a critical barrier to adequate medication management in the context of living alone.[62] While the importance of social determinants of health, including living arrangements, is recognized,[73] clinicians remain unsure how to intervene.[3,74,75] Detecting cognitive impairment in older adults who live alone should spur efforts by clinicians and health systems to identify medication-related challenges; achieve feasible, goal-concordant regimens through deprescribing to reduce unnecessary polypharmacy[41]; and engage noncohabiting distance caregivers, if available.[11,76] A crucial, practical next step will be to standardize this information, including residential status (eg, living alone or other relevant information, such as residing in an assisted living facility) and cognitive status (eg, for those with mild cognitive impairment or dementia), in discrete electronic health record fields, making it viewable to the entire care team.[48-50,73] An additional promising avenue would be increasing health system support for pharmacist-led medication reviews.[52,53]

Findings regarding the formidable barriers that participants experienced navigating health care–related bureaucracy underscore the importance of expanding access and bolstering home- and community-based services (HCBS). Medication management issues voiced by participants are a microcosm of challenges faced by the US health care system at large, characterized by heavy dependence on medications, with insufficient investment in HCBS to support oversight and adherence.[77,78] For older adults living alone with cognitive impairment who qualify for Medicaid, enhanced medication management support could occur through increased support for home care aides or assistive devices that facilitate adherence. However, many (79%) older adults living alone with cognitive impairment do not qualify for Medicaid and instead rely on scattered, informal, unpaid support.[4,51] Therefore, innovations are needed to expand Medicaid coverage and improve access to HCBS. Additionally, state regulations vary considerably in terms of the scope of practice of home care aides in medication management; more research is needed to elucidate whether changes in home care aide scope of practice could help in meeting the needs of older adults living alone with cognitive impairment.[79] The Centers for Medicare & Medicaid Services' Guiding an Improved Dementia Experience (GUIDE) model represents an avenue for medication-related support,[80] but guidance specifically related to older adults living alone with cognitive impairment is needed for the GUIDE Model to be successful in serving this population.[5,55,56]

Limitations

While we interviewed participants with varying degrees of social engagement, participants likely did not represent the most socially isolated individuals whose recruitment into research studies is challenging.[81,82] While participants resided in 3 geographically disparate states, most were in California, and the results may not represent the experiences of individuals from other regions. We did not systematically collect information regarding the total number of prescribers for each participant or the frequency or mode of contact between participants and social contacts. Additionally, our protocol was not designed to disaggregate the role of factors such as limited English proficiency compared with living alone status in our findings. Finally, the depth of qualitative findings was constrained to varying degrees by participants' cognitive impairment and language barriers; narratives may have been skewed toward those with milder cognitive impairment and/or higher English proficiency.

Conclusions

This large qualitative study including diverse participants from 3 states highlights the barriers older adults living alone with cognitive impairment experience while managing their medications. Targeted interventions are needed to support medication management in this population, starting with more systematic integration of living alone status into electronic health records.

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